



MERGED GROUP 2

GROUP 2

DE GUZMAN, Jan Denver
PASCUAL, Jasmine Nicole
MEDENILLA, Jose Anton
TIÑANA, Chelzy Gracel
VERGARA, Revie Shane

GROUP 5

ABAMONGA, Michelle Sophia
BAROLA, Francesca Denise
LOPEZ, Elijah
MANANGUIT, Jared Asher
NACU, Francesca Ellaiza

Section: **3ECE-B**

Date: December 7, 2025

ALU DESIGN

8-Bit Arithmetic and Logic Unit (ALU) using Verilog

I. SCREENSHOTS OF WAVEFORMS

First Setup

<pre>rst = 1 clk = 10 ns opcode_0 = 1000 ns opcode_1 = 500 ns opcode_2 = 250 ns</pre>	<pre>data_in_A = Random at Fixed Interval = 1000 ns data_in_B = Random at Fixed Interval = 1000 ns shift_mode_A = 0 shift_mode_B = 0</pre>
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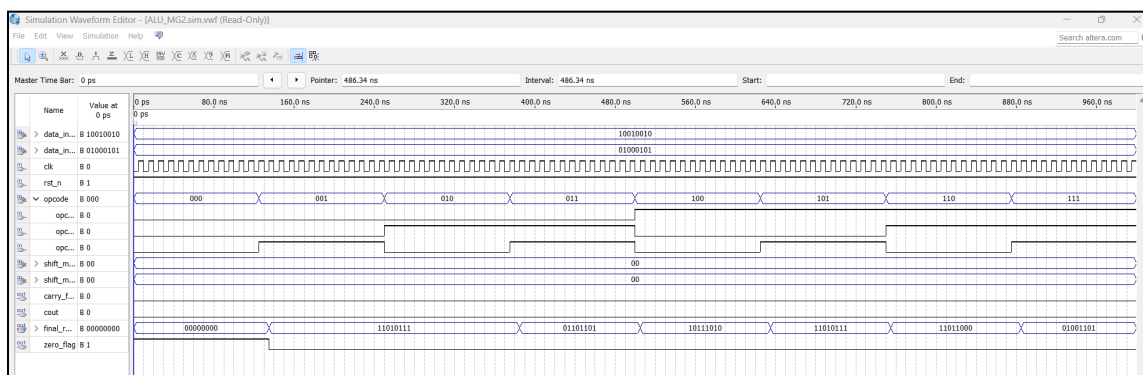


Figure 1 - Verilog Waveform of the Setup 1

Second Setup

rst = 1	data_in_A = 1
clk = 10 ns	data_in_B = 1
opcode_0 = 1000 ns	shift_mode_A = 0
opcode_1 = 500 ns	shift_mode_B = 0
opcode_2 = 250 ns	

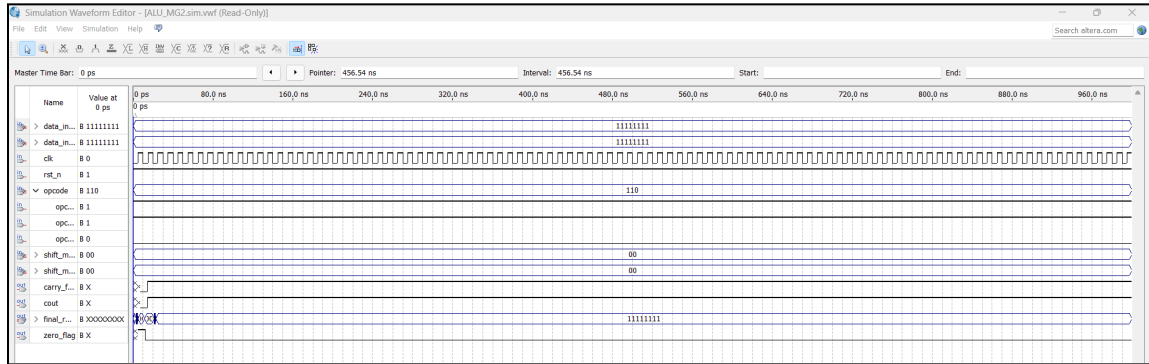


Figure 2 - Verilog Waveform of the Setup 2

Shift Register Simulation

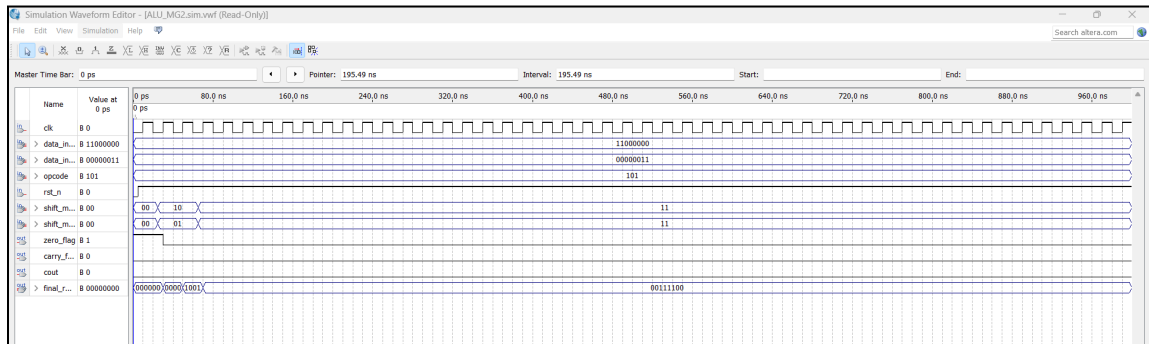


Figure 3 - Verilog Waveform of Shift Register

II. LIST OF CONTRIBUTIONS

GROUP MEMBERS	TASK ACCOMPLISHED
ABAMONGA, Michelle Sophia	Experiment 7 - Arithmetic Unit
BAROLA, Francesca Denise	Experiment 7 - Arithmetic Unit
DE GUZMAN, Jan Denver	Experiment 8 - Registers
LOPEZ, Elijah David	Experiment 7 - Arithmetic Unit ALU Design - Opcodes
MANANGUIT, Jared Asher	Experiment 7 - Arithmetic Unit ALU Design - Opcodes
MEDENILLA, Jose Anton	Experiment 8 - Registers
NACU, Francesca Ellaiza	Experiment 7 - Arithmetic Unit
PASCUAL, Jasmine Nicole	Experiment 8 - Registers
TIÑANA, Chelzy Gracel	Experiment 8 - Registers ALU Design - Summary of Waveforms
VERGARA, Revie Shane	Experiment 8 - Registers ALU Design - Waveforms Simulation